

4.1 A 2100-lb tractor is used to lift 900 lb of gravel. Determine the reaction at each of the two (a) rear wheels A, (b) front wheels B.

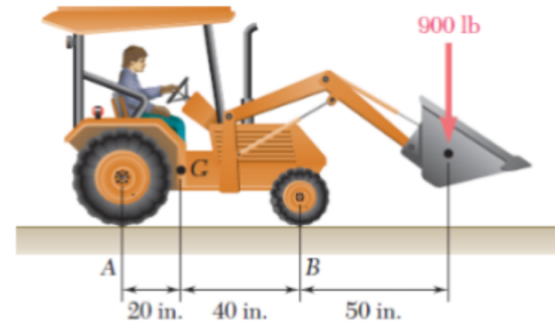
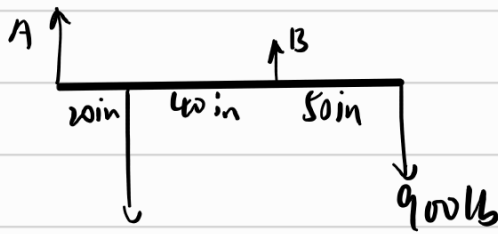


Fig. P4.1

a) $\sum M_B = 0$

$$2100(40) - 900(50) - 2(R_A)(60) = 0$$

$$120R_A = 39000$$

$$R_A = +325 \text{ lb}$$

b) $\sum M_A = 0$

$$-2100(20) - 900(110) + 2(R_B)(60) = 0$$

$$R_B = +1175 \text{ lb}$$

Checking

$$\sum F = 0 \Rightarrow 2(R_A + R_B) - 2100 - 900 = 2(325 + 1175) - 3000 = 3000 - 3000 = 0 \text{ (checked)}$$

4.55 A collar B of weight W can move freely along the vertical rod shown. The constant of the spring is k , and the spring is unstretched when $\theta = 0$. (a) Derive an equation in θ , W , k , and l that must be satisfied when the collar is in equilibrium. (b) Knowing that $W = 300 \text{ N}$, $l = 500 \text{ mm}$, and $k = 800 \text{ N/m}$, determine the value of θ corresponding to equilibrium.

$$a) F=kx \quad kx \sin \theta - W = 0$$

$$l_1 \cos \theta = l$$

$$l_1 = \frac{l}{\cos \theta}$$

$$x = \frac{l}{\cos \theta} - l$$

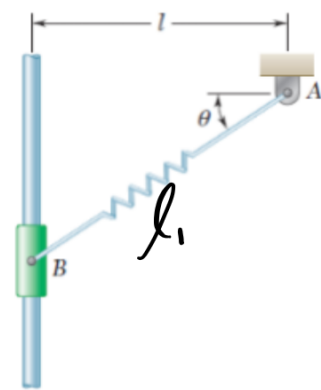
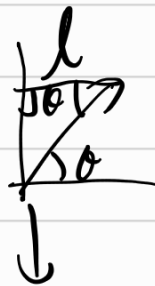


Fig. P4.55

$$k\left(\frac{l}{\cos \theta} - l\right) \sin \theta - W = 0$$

$$kl\left(\frac{1}{\cos \theta} - 1\right) \sin \theta = W$$

$$\tan \theta - \sin \theta = \frac{W}{kl}$$

$$b) \tan \theta - \sin \theta = \frac{300}{800(0.5)} = \frac{3}{4}$$

$$\text{let } t = \tan \frac{\theta}{2}$$

$$\frac{2t}{1-t^2} - \frac{2t}{1+t^2} = \frac{3}{4}$$

$$\frac{2t(1+t^2) - 2t(1-t^2)}{1-t^4} = \frac{3}{4}$$

$$4(2t + 2t^3 - 2t + 2t^3) = 3(1-t^4)$$

$$16t^3 = 3 - 3t^4$$

$$3t^4 + 16t^3 - 3 = 0$$

$$t = 0.5538$$

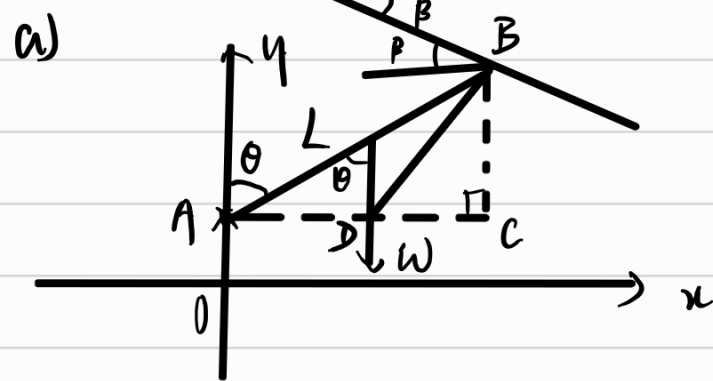
$$t = -5.34 \text{ (reject)}$$

$$\tan \frac{\theta}{2} = 0.5538$$

$$\frac{\theta}{2} = 28.98^\circ$$

$$\theta = 57.96^\circ \approx 58^\circ$$

4.85 An 8-kg slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium and that $\beta = 30^\circ$, determine (a) the angle θ that the rod forms with the vertical, (b) the reactions at A and B.



$$\begin{aligned}
 BC &= L \cos \theta & \tan \beta &= \frac{DC}{BC} \\
 AC &= L \sin \theta & \tan 30^\circ &= \frac{\frac{1}{2} L \sin \theta}{\frac{1}{2} L \cos \theta} \\
 DC &= \frac{1}{2} AC = \frac{1}{2} L \sin \theta & \frac{1}{\sqrt{3}} &= \frac{1}{2} \tan \theta \\
 & & \tan \theta &= \frac{2}{\sqrt{3}} \quad \theta = 49.11^\circ
 \end{aligned}$$

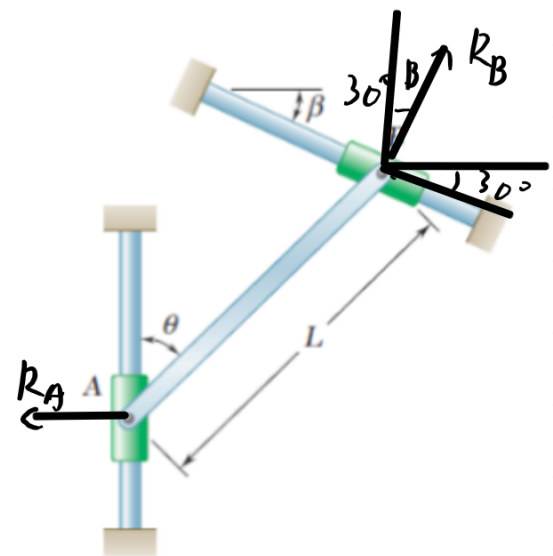


Fig. P4.84 and P4.85

b) $\sum F_y = 0$ $R_B \cos 30^\circ - W = 0$ $\frac{\sqrt{3}}{2} R_B = W = mg = 8(9.81) \text{ N}$

$\sum F_x = 0$ $R_B \sin 30^\circ - R_A = 0$ $R_B = 90.6209 \text{ N}$

$\frac{1}{2} R_B - R_A = 0$

$\frac{1}{2} (90.6209) - R_A = 0$

$R_A = \frac{1}{2} (90.6209) = 45.3105 \text{ N}$

4.135 The 50-kg plate $ABCD$ is supported by hinges along edge AB and by wire CE . Knowing that the plate is uniform, determine the tension in the wire.

$$\begin{aligned}
 \vec{CE} &= -240\hat{i} + 600\hat{j} - 400\hat{k} \\
 |\vec{CE}| &= \sqrt{(-240)^2 + 600^2 + (-400)^2} = 760 \text{ (mm)}
 \end{aligned}$$

$$W = 50(9.81) = 490.5 \text{ N}$$

The force acting on $CE = \vec{F}_{CE} = T \left(\frac{\vec{CE}}{|\vec{CE}|} \right)$

$$= \frac{T}{760} (-240\hat{i} + 600\hat{j} - 400\hat{k})$$

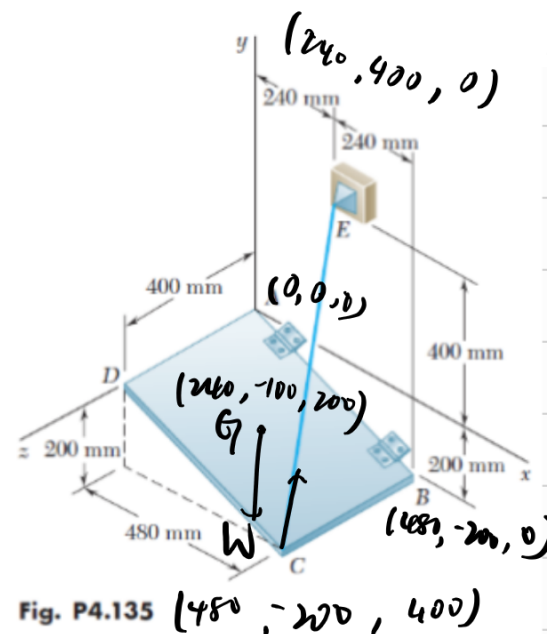


Fig. P4.135

$$\begin{aligned}
 \vec{AB} &= 480\hat{i} - 200\hat{j} \\
 \lambda_{AB} &= \frac{\vec{AB}}{|\vec{AB}|} = \frac{480\hat{i} - 200\hat{j}}{\sqrt{(480)^2 + (-200)^2}} = \frac{480\hat{i} - 200\hat{j}}{520}
 \end{aligned}$$

$$\vec{AC} = 480\vec{i} - 200\vec{j} + 400\vec{k}$$

$$\vec{AG} = 240\vec{i} - 100\vec{j} + 200\vec{k}$$

$$\sum M_{AB} = 0 \quad \vec{\lambda}_{AB} \cdot (\vec{AC} \times \vec{F}_{CE}) + \vec{\lambda}_{AB} \cdot (\vec{AG} \times \vec{W}) = 0$$

$$\left(\frac{12}{13}\vec{i} - \frac{5}{13}\vec{j}\right) \cdot \left(\frac{T}{760} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 480 & -200 & 400 \\ -240 & 600 & -400 \end{vmatrix}\right) + \left(\frac{12}{13}\vec{i} - \frac{5}{13}\vec{j}\right) \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 240 & -100 & 200 \\ 0 & -W & 0 \end{vmatrix} = 0$$

$$\frac{T}{13 \times 760} (12\vec{i} - 5\vec{j}) \cdot (160000\vec{i} + 96000\vec{j} + 240000\vec{k}) + \frac{12}{13} (200W) = 0$$

$$\frac{T}{760} (-2400000) + 2400W = 0$$

$$T = \frac{-240000W}{-2400000} (760) = 0.76W = 0.76(490 \cdot 5) \text{ N}$$

$$\underline{\underline{= 372.78 \text{ N}}}$$